SRILAKSHMI SESHADRI

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PROFESSIONAL SUMMARY

Final year CS Engineering student (CGPA 8.92) with hands-on experience in Machine Learning, Computer Vision, and full-stack AI deployment. Built production-ready ML systems achieving R^2=0.96 and 95%+ accuracy. Developed a Transformer-based sports analytics system using MediaPipe, PyTorch, and FastAPI. TCS CodeVita Global Rank 7917.

TECHNICAL SKILLS

Languages: Python, Java, C++
ML / AI: PyTorch, scikit-learn, XGBoost, MediaPipe, joblib, StandardScaler, Transformers (ViTPose)
Web & Frameworks: FastAPI, React, TypeScript, Streamlit
Data: pandas, NumPy, matplotlib, seaborn
Tools: Git, GitHub, Google Colab, Jupyter Notebook
Concepts: ML pipelines, Computer Vision, Pose Estimation, EDA, Feature Engineering, Real-time Inference

PROJECTS

Transformer-Based Pose & Performance Analysis of Basketball Using AI

- Built an AI sports analytics system extracting 33 skeletal keypoints via MediaPipe and analyzing joint relationships using a ViTPose Transformer model to classify shooting, dribbling, and defensive stances.
- Developed a 0–100% performance scoring engine with joint-level corrective feedback and temporal analysis for movement consistency across video frames.
- Architected a full-stack system with React + TypeScript frontend and FastAPI + PyTorch backend, exposing REST API endpoints for image uploads, video processing, and pose retrieval.
- Implemented a session tracking module enabling players to monitor performance progression over time.

Healthcare Prediction Web Application

- Built an AI-powered healthcare diagnostic tool using Streamlit and Python, reducing manual data entry time by 40% on simulated patient datasets.
- Designed scalable backend modules for data processing and real-time inference, reducing system latency by 20%.
- Delivered end-to-end solution covering ML model integration, web-app development, and AI deployment.

Wind Turbine Power Output Prediction

- Developed a regression pipeline on 33+ environmental and operational variables using XGBoost and Random Forest, achieving R^2=0.96, MAE=0.012 kW, and RMSE=0.018 kW.
- Performed feature engineering and EDA (correlation heatmap, feature importance) to identify wind speed and generator RPM as dominant predictors.
- Built a real-time inference pipeline using joblib for model persistence and deployment, integrable with live wind farm monitoring systems.

CTR Prediction Using Logistic Regression Trained a logistic regression classifier on user behavioral data to predict ad-click probability, achieving 96% accuracy and AUC of 0.99.

- Applied feature engineering from timestamps (Hour, DayOfWeek), MinMaxScaler normalization, and visualized performance via confusion matrix and ROC curve.

CERTIFICATIONS & ACHIEVEMENTS

- **Generative AI Job Simulation** — Forage
- **Introduction to Generative AI Studio** — Simplilearn SkillUp
- **AI Training & Internship** — Ediglobe x Kshitij
- **TCS CodeVita Season 13** — Global Rank 7917

EDUCATION

B.E. in Computer Science & Engineering 2023 – 2027
Sapthagiri College of Engineering, VTU | CGPA: 8.92

Pre-University (2nd PUC) 2021 – 2023
Vidya Mandira Pre University College | 89.66%

10th Standard (CICSE) 2020 – 2021
Cluny Convent High School | 86.4%